

AmirPouya Moeini

Ph.D. Student @ University of Cambridge, Trinity Hall College

✉ sam297@cam.ac.uk [in](https://www.linkedin.com/in/pouya-moeini) [pouya-moeini](https://www.github.com/pouyamoeini) pouyamoeini.github.io

Research Interests

I am broadly interested in **high-dimensional probability**, **information theory**, and **combinatorics**. My PhD studies error-exponent limits in information theory (achievability and converse), using tools from probabilistic combinatorics, with a particular focus on mismatched and universal settings.

Education

Ph.D. in Engineering

📅 Oct 2023 – Ongoing

🏛️ University of Cambridge, UK — Probabilistic Systems, Information, and Inference Group (PSI²)

Supervisor: Prof. Albert Guillén i Fàbregas

B.Sc. in Computer Engineering (Minor in Applied Mathematics)

📅 Sep 2018 – Jun 2023

🏛️ Sharif University of Technology, Iran

Thesis: *Information-theoretic bounds for online learning* (20/20)

Publications

“Two-Class Joint Source-Channel Coding: Expurgated Exponents with i.i.d. Distributions”

[\[Link\]](#)

Moeini, A. and Guillén i Fàbregas, A., 2026 International Zurich Seminar on Information and Communication (IZS).

“Dual-Domain Exponent of Maximum Mutual Information Decoding”

[\[Link\]](#)

Moeini, A. and Guillén i Fàbregas, A., 2025 IEEE Information Theory Workshop (ITW).

“Class-Based Expurgation Attains Csiszár’s Expurgated Source-Channel Exponent”

[\[Link\]](#)

Moeini, A. and Guillén i Fàbregas, A., 2025 IEEE Information Theory Workshop (ITW).

Academic Visits

🏛️ **National University of Singapore**, Singapore

October 2025

School of Computing – Institute of Data Science, visiting Dr. Jonathan Scarlett

🏛️ **University of Brescia**, Brescia, Italy

September 2025

Department of Information Engineering, visiting Dr. Marco Dalai

Honours and Awards

2023 **Cambridge International Trust – Full Scholarship** ($\approx$ £62,000 per year)

Cambridge, UK

2022 **Summer AScI @ Aalto University Fellowship**

Espoo, Finland

2018 **Gold Medal** *12th International Olympiad on Astronomy and Astrophysics (IOAA)*

Beijing, China

2017 **Gold Medal** *13th National Astronomy and Astrophysics Olympiad*

Iran

Teaching Experience

Supervision 🏛️ University of Cambridge

- Information Theory and Coding (3F7), Michaelmas 24, 25
- Data Transmission (3F4), Lent 25, 26

Teaching Assistant 🏛️ Sharif University of Technology

- Information theory & Coding, Fall 22
- Theoretical Machine learning, Fall 22
- Statistical Machine learning, Spring 22
- Mathematical Statistics, Fall 21
- Data Structure & Algorithm, Spring 21
- Linear algebra, Spring 21
- Regression Analysis, Spring 21
- Artificial intelligence, Fall 21, Spring 22

Selected Courses

University of Cambridge

Computational statistics and machine learning, Advanced information theory and coding

Sharif University of Technology

Probability

Information theory and coding, Information-theoretic methods in high-dimensional statistics, High-dimensional probability, Graphical models, Stochastic processes, Bayesian methods in statistics

Mathematics

Real analysis, Algebra, Linear algebra, Applied linear algebra, Probability & statistics

Machine Learning

Machine learning theory, Artificial intelligence, Computer simulation, Modern information retrieval

Computer Science

Design algorithms, Data structures and algorithms, Compiler design, Database Design

Audited Courses

Probabilistic combinatorics, Entropy methods in combinatorics, Ramsey theory, Random discrete structures (*Cambridge*)

Research Experience

Approximate Bayesian Computation in the Presence of Outliers

 Jul 2022 – Oct 2022

- Probabilistic Machine Learning Group, Aalto University, Finland
- Supervisor: *Prof. Samuel Kaski*.

- Developing computationally efficient algorithms that enhance the robustness of Approximate Bayesian Computation (ABC) inference under outliers and model misspecification.

Information-Theoretic Bounds for Online Learning Problems

 Oct 2021 – Mar 2023

- Machine Learning Laboratory, Sharif University of Technology, Iran
- Supervisors: *Dr. Mahdieh Soleymani* and *Dr. Hassan Hafez-Kolahi*.

- We introduce information-theoretic tools for online learning with adversarial data and derive performance upper bounds via conditional mutual information.

Skills

Programming Language: Python (*advanced*), Java (*advanced*), C / C++, SQL.

Statistical Programming Languages: R (*advanced*), Julia (*advanced*), Stata, MatLab & Octave.

Tools and Software: MS Excel, L^AT_EX, Jupyter NB / G Colab, PyTorch, Git / GitHub

Language: Farsi/Persian (*native*), English (*fluent*), French (*basic*).